

Vyacheslav Kurnevich

MIPT DREC, 2nd year

Looking for an internship in the field of system programming

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ACADEMIC PROJECTS, 2ND YEAR

•DSL for RISC-V-32IM

generation of interpreter, decoder and tests based on Ruby-based ISA description

- Ruby, C++, Python
- add'l tools: ELFIO, QEMU for test generation

•Toy-ISA Interpreter

interpreter with optimization via binary translation of "hot" code blocks to assembly

- tools: C, C++, asmc library, Ruby

•Chat using POSIX real-time signals, my_bash, my_cp

notable assignments from programming workshops

- tools: C, system calls

ACADEMIC PROJECTS, 1ST YEAR

•Compiler

compiler for a custom programming language

- syntax parsing via recursive descent, AST construction
- translation of AST nodes and symbol table to x86-64 assembly
- tools: C, dot, make

•Hash Table Data Lookup Optimization

- implementation of a hash table with various hash functions
- SSE2 \AVX \AVX2 intrinsics
- Profiling, Optimization (achieved 2.37x speedup)
- tools: C, make, asm, perf

•Mandelbrot Set Renderer Optimization

- implementation of algorithm for rendering the Mandelbrot set
- SFML, SSE2 \AVX \AVX2 intrinsics
- Profiling, Optimization (achieved 6.8x speedup)
- tools: C, make, perf

•Differentiator

program performs differentiation of single-variable functions

- implementation of tree data structure
- parsing of input function using recursive descent
- dump of differentiated expression to tex file
- tools: C, dot, make

•Akinator

custom implementation of the Akinator game

- implementation of tree data structure
- saving a database of characters known to the game
- adding new characters to the database
- tools: C, dot, make

TECHNICAL SKILLS

Systems & low-level: C, C++, x86-64 asm, SIMD (SSE2/AVX/AVX2), asmc, perf, ELFIO, QEMU, LLVM (snippy)

Scripting: Python, Ruby

Tooling: Git, Make, CMake, IDA, Graphviz/dot, LaTeX